

Siqi Guo

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Summary: CMU M.S. ECE graduate with expertise in robot perception/localization, embedded control, bio-inspired locomotion, and autonomous planning. Currently developing **LiDAR-inertial SLAM/localization** and parallelized **C++ IK and motion planning pipelines** for wheeled humanoid warehouse robots at Yondu AI. Previously embedded software engineer at Cepton (LiDAR systems) and research assistant in the CMU Biorobotics Lab (**EigenBot**, **MMPUG**). Published in ICRA 2025, AAAI 2023, and IEEE Neural Networks. Seeking research collaboration in Embodied AI systems, world models for robotics, real-time robot learning, and scalable deployment of foundation models on physical platforms.

EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Master of Science in Electrical and Computer Engineering; GPA: 3.91/4.0

May 2025

Relevant Courses: Foundations of Computer Systems, Embedded System Software Engineering, Machine Learning Signal Processing, Distributed Systems, Autonomous Control System, Storage Systems, Modern Computer Architecture and Design, Large-scale Distributed Machine Learning and Optimization, Introduction to Embedded Systems, Multi-Robot Planning and Coordination, How to Write Fast Code II, Software Development for Computational Science and Engineering

Tianjin University

Tianjin, China

Bachelor of Science in Computer Science and Technology; GPA: 3.84/4.0

June 2023

Relevant Courses: Data Structure, Operating System, Computer Networks, Principles of Database, Parallel Computing, Natural Language Processing, Pattern Recognition and Deep Learning

PUBLICATIONS

- Zhang, Z., **Guo, S.**, et al. “Bio-inspired Distributed Neural Locomotion Controller (D-NLC) for Robust Locomotion and Emergent Behaviors” (May '25). *2025 IEEE International Conference on Robotics and Automation (ICRA 2025)*.
- Bingdao, F., Di, J., et al. “Backdoor attacks on unsupervised graph representation learning” (August '24). *Journal of Neural Networks* (Vol. 180, p.106668).
- Jin, D., Feng, B., **Guo, S.**, et al. “Local-Global Defense Against Unsupervised Adversarial Attacks on Graphs” (June '23). In the *Thirty-Seventh AAAI Conference on Artificial Intelligence* (Vol. 37, No. 7).
- Ding, Y., & **Guo, S.** “Conditional generative adversarial networks: Introduction and application” (November '22). In the *2nd International Conference on Artificial Intelligence, Automation, and High-Performance Computing (AIAHPC 2022)* (Vol. 12348, pp. 258-266). SPIE.

RESEARCH & WORK EXPERIENCE

Yondu AI | Y Combinator, Early-stage Humanoid Robotics Startup

Los Angeles, CA

Robot Intern (Full-time)

Feb 2026 - Present

- Parallelized Motion Planning for Wheeled Humanoid Warehouse Robots
 - Developing parallelized **C++ inverse kinematics (IK) and motion planning pipelines** for wheeled humanoid warehouse robots, targeting real-time performance for autonomous manipulation tasks.
 - Designed **experience-based roadmap cache** system with persistence to accelerate similar pose/trajectory queries, reducing IK solving time for repeated warehouse tasks.
 - Developed a warehouse cart scanning simulation pipeline with QR code localization, IK dispatch for barcode poses, and live virtual camera feedback during scanning.
- Ouster Integration, Dynamic IMU Estimation, and LiDAR-Inertial SLAM
 - Resolved persistent **asymmetric yaw drift and odometry snapping** for Ouster odometry through **extrinsic calibration** (KISS-ICP), EKF fusion tuning, **dynamic IMU bias correction**, and **point cloud deskewing**.
 - Reproduced **Fast-LIO2** and **LIO-SAM** across Livox, Ouster, Velodyne, and internal Ouster datasets, generating clean PCD maps and implementing **Fast-LIO2 localization** for Ouster-based robot operation beyond prior 2D SLAM Toolbox workflows.
- Data Collection and Post-processing Pipeline Optimization & Acceleration
 - Redesigned cloud-to-dataset post-processing pipeline, achieving **5-7x speedup** and improving throughput from **28 to 145 FPS**; reduced cloud transfer by **92%** via parallel segment download and coverage-aware data selection.
- Warehouse Robot Field Deployment and Patrol System
 - Deployed robots at **GPA Logistics** and **Cainiao** warehouses, handling site network configuration, production bring-up, field debugging, and GPA patrol replay/coverage benchmarking.

Cepton, Inc. | LiDAR Technology Company

San Jose, CA

Embedded Software Engineer (promoted from Software Engineer Intern, Full-time)

May 2025 - Jan 2026

- MST (Magnosteer Control) System Development
 - Developed embedded **PD control with feedforward** for precision mirror movement control in LiDAR systems, implementing encoder auto-calibration and trajectory interpolation in **C/C++**.

- * Optimized embedded system architecture by redesigning **SPI transfers and event queues**, achieving **73% CPU utilization reduction** (90% → 24%) on ARM-based controllers.
- * Built comprehensive **control simulation framework** in C and Python for algorithm validation, enabling systematic benchmarking of PD vs. PD-with-feedforward control strategies.
- * Engineered real-time **MST GUI and debug logging system** in **Rust**, providing field engineers with live system monitoring and parameter tuning capabilities.
- * Delivered **72% power consumption reduction** (2.9W → 0.8W) on Ultra B3.3 120° LiDAR units through embedded firmware optimization and Python-based power analysis tools.

Eigenbot - Bio-inspired Distributed Control for Modular Robot

Biorobotics Lab Research Assistant (Part-time)

Advisor: Professor Howie Choset, Research Scientist Lu Li

Pittsburgh, PA

Dec 2023 - Oct 2024

Carnegie Mellon University

- Modular Robot Bio-inspired Curve Walking Implementation
 - * Designed and implemented a bio-inspired curve walking algorithm in CoppeliaSim for hexapod locomotion, optimizing controller weights for smooth gait transitions and adding steering control into Eigenbot's behavior tree.
- Eigenbot Firmware Development and Testing
 - * Developed Eigenbot EEPROM parameters auto-setting tools, which accelerated Hexapod gaits tuning process.
 - * Tested the firmware. Redesigned and debugged the module communication protocol to ensure the robot's stability and reliability. Conducted the experiments with Logan, analyzed the gaits and fine-tuned firmware parameters.
 - * Integrated Foot Sensor Feedback into the robot's message protocol.
 - * Collected EigenBot's data from the real-world using OptiTrack, developing a pipeline to assess metrics like Gait Phase Difference and Leg Stance Duration, benchmarking its performance with centralized predefined gait.

Theory and Applications of Graph Convolutional Network

Research Assistant (Part-time)

Advisor: Associate Professor Jin, Di

Tianjin, China

Nov 2021 - May 2023

Tianjin University

- Studied unsupervised graph representation learning and robustness by analyzing GAT/GCN behavior within Deep Graph Infomax (DGI), integrating mutual information maximization, and benchmarking adversarial defenses against Metattack and Netstack, including RoSA, Pro-GNN, PA-GNN, GraphCL, and GraphSS.
- Designed a novel **unsupervised backdoor attack** on GNNs based on GIN and Momentum Contrast, formulated the threat model, derived the loss function, and implemented trigger injection; achieved higher Attack Success Rate and lower Clean Accuracy Drop than prior unsupervised graph attacks.

PROJECTS

CloudFS, a Deduplicated Cloud File System | CMU 18-746 Course Project Oct 2024 - Dec 2024

- Implemented CloudFS, leveraging the properties of local SSD and cloud storage to make data-placement decisions.
- Identified and eliminated duplicate data with Rabin fingerprinting. Minimized storage usage and optimized data transfer in the form of segments. Designed buffer file to manually handle the file data transfer.
- Implemented the IOCTL functions to support snapshot operations, such as create, restore, install, and uninstall.
- Added in-memory LRU caching to further improve performance and reduce cloud operation costs, reaching **NO.1** on the scoreboard among all students.

BusTub, a Relational DBMS | Personal Project based on CMU 15-645 May 2024 - July 2024

- Implemented DBMS storage layer (LRU-K, disk scheduler, buffer pool, extendible hash index), query execution (operator executors, optimizer rules), and transaction support with optimistic multi-version concurrency control.

Distributed File System Design | CMU 15-640 Course Project Feb 2024 - Apr 2024

- Built a File-Caching Proxy over RMI with open-close session semantics, LRU cache replacement, and version-based freshness control; implemented low-level RPC serialization for NFS operations.
- Designed a Two-phase Commit Protocol with failure recovery and lost-message handling; tuned a 3-tier auto-scaling web service and benchmarked system bottlenecks under dynamic load.

Blind Source Signal Decomposition & Analysis | CMU 18797 Research Project Oct 2023 - Dec 2023

- Filtered the mixed HDEMG signals and detected peaks to slice the peak-wave windows. Used the PCA algorithm to reduce the dimensions of the data and K-Means to cluster the potentially same Motor Units' signals.
- Based on the clustered results, applied DTW, Covariance, Cosine in a multi-threshold way to compare the wave similarity. Verified the firing instances for each Motor Unit based on the comparison results.

SKILLS

Programming: C/C++, Python, Rust, LaTeX, Verilog

Frameworks: ROS, TensorFlow/PyTorch, Isaac Sim, Drake

Tools: GIT, Docker, MySQL, VSCode, Gem5, PSoC, Vitis

Platforms: MacOS, Linux, Arduino, GCP